Advanced Web Technology

23CP306T

Semester: 5

Branch: B.Tech. CSE

Subject Name/Lab Name					Advanced Web Technology Theory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	1	0	1	1	25	50	25	-	-	100

UNIT 1 Web Development with Node and Express The JavaScript Revolution, Introducing Express, A Brief History of Express, The Node Ecosystem, What is Express.js, How Express.js Works, MVC Structure and Modules	5 Hrs.
UNIT 2 React React's Past and Future, How React Works, React State Management	4 Hrs.
UNIT 3 Django Intro, MVT Architecture, Views, URL, Templates, Models, Database Connectivity (PostgreSQL)	5 Hrs.
UNIT 4 Flask Introduction, Flask- Application, Routing, Variable Rules, URL Building, HTTP methods, Templates, SQLAlchemy	3 Hrs.

COURSE OUTCOMES

- **CO1-** Explain the concepts of Web Application
- **CO2-** Identify and explain various components of Web Application stack
- **CO3-** Illuminate technologies and tools involved in Web Application Development
- **CO4-** Understand and Identify the use of Web Services in Web Programming
- **CO5-** Learn JavaScript based runtime environments: NodeJS, ReactJS.
- **CO6-** Understand Python frameworks for web development: Django and Flask

Pillars of Web Development

- Essentially, just three things: **HTML, CSS** and **JavaScript**—the three pillars of the web.
  - Together, these three pillars make every website work,
  - defining the content to be displayed (HTML),
  - telling a browser how to display that content (CSS),
  - and making the content interactive and modelling its behavior (JavaScript), respectively.

Create a single webpage that uses HTML, CSS and JavaScript

Index.html

```html
<head>
<title>Single Slide Example</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <div class="slide">
    <h1 id="title">Welcome to My Slide</h1>
    <p id="text">This is a simple one-slide example.</p>
    <button onclick="changeContent()">Change Content</button>
  </div>

  <script src="script.js"></script>
</body>
```

script.js

```javascript
function changeContent() {
  document.getElementById("title").innerText = "Slide Updated!";
  document.getElementById("text").innerText =
    "The content of this single slide was changed using JavaScript.";
}
```

style.css

```css
body {
  font-family: Arial, sans-serif;
  background: #f4f6f8;
  display: flex;
  justify-content: center;
  align-items: center;
  height: 100vh;  }
button {
  margin-top: 15px;
  padding: 10px 15px;
  border: none;
  background: #007bff;
  color: white;
  border-radius: 5px;
  cursor: pointer;        }
.slide {
  background: white;
  padding: 30px;
  width: 400px;
  text-align: center;
  border-radius: 10px;
  box-shadow: 0 4px 10px
         rgba(0, 0, 0, 0.1);}
```

Purpose of JavaScript

- JavaScript is a scripting language used to develop web pages. Developed in Netscape, JS allows developers to create a dynamic and interactive web page.
- Applications like Google Maps. Powered by JavaScript, which interacts with the browser without communicating with the servers.
- Node.js allows developers to use JavaScript to create a web server.

Various other applications of JavaScript include:

- Games (EaselJS library)
- Art (Canvas element of HTML5 in JavaScript)
- Small robots, creative maker projects, and IoT devices (Node.js ecosystem)
- Smart Watches (an example is Pebble.js) and more

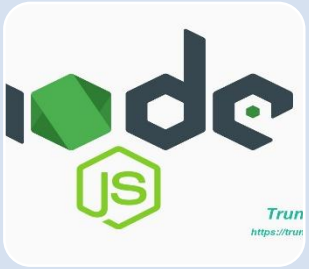
The JavaScript Revolution

- From Browser to Everywhere
  - 1995: Created by Brendan Eich (Netscape)
  - 2009: Node.js enables JavaScript on the server
  - 2010s: Full-stack JavaScript development
- Why JavaScript Took Over
  - Single language for frontend/backend
  - NPM ecosystem (largest package registry)
  - Event-driven, non-blocking I/O

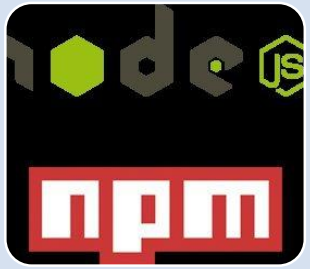
Evolution of JavaScript Timeline

1995: JavaScript created

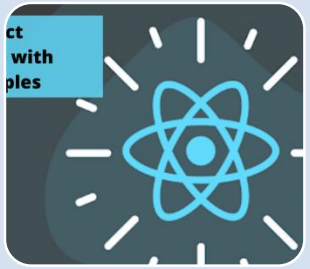
2005: AJAX revolution (Gmail, Google Maps)



2009: Node.js released



2010: NPM created



2013: React introduced

2015: ES6 Modern JavaScript

2016+: Full-stack JavaScript dominance

What is AJAX?

- **AJAX** = Asynchronous JavaScript and XML
- **Not a single technology** but a combination of:
  - HTML/CSS (for presentation)
  - DOM (for dynamic content display)
  - JSON/XML (for data interchange)
  - XMLHttpRequest object (for asynchronous communication)
  - JavaScript (to bring it all together)

What is NPM?

- **NPM** = Node Package Manager (but now stands for "npm" itself)
- **Three distinct components:**
  - **Registry** - Public database of JavaScript packages
  - **Website** - npmjs.com for discovering packages
  - **CLI Tool** - Command-line interface for managing packages
- **Statistics (as of 2024):**
  - Over 2.5 million packages
  - 75+ billion downloads monthly
  - 17+ million developers

Advantages of JavaScript

1. Client-Side Execution

JavaScript runs directly in the user's browser.

- Reduces load on the server
- Faster response time
- Improves overall performance of web applications

👉 Example: Form validation without sending data to the server.

2. Faster Web Applications

Since JavaScript executes on the client side:

- No need to wait for server responses
- Immediate feedback to users
- Enhances user experience

👉 Example: Real-time input validation and UI updates.

3. Easy to Learn and Use

JavaScript syntax is:

- Simple and readable
- Similar to C, Java, and Python
- Beginner-friendly

👉 Many beginners start web development with JavaScript.

4. Supports Dynamic Content

JavaScript allows web pages to be interactive and dynamic.

- Change HTML and CSS at runtime
- Update content without refreshing the page
- Create animations and interactive elements

👉 Example: Image sliders, dropdown menus, pop-ups.

5. Platform Independent

JavaScript runs on:

- Windows
- Linux
- macOS
- Mobile devices

👉 Only a web browser is required.

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6. Wide Browser Support

All modern browsers support JavaScript:

- Chrome
- Firefox
- Edge
- Safari

👉 No need for additional plugins.

7. Supports Asynchronous Programming

JavaScript supports asynchronous operations using:

- Callbacks
- Promises
- async/await

Benefits:

- Non-blocking execution
- Better performance
- Smooth user experience

👉 Example: Fetching data from an API in the background.

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8. Large Ecosystem and Libraries

JavaScript has a rich ecosystem:

- Frameworks: React, Angular, Vue
- Libraries: jQuery, D3.js
- Package manager: npm

👉 Speeds up development and reduces effort.

9. Full-Stack Development

With Node.js, JavaScript can be used for:

- Frontend development
- Backend development
- API creation

👉 Same language for client and server reduces complexity.

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10. Supports Event-Driven Programming

JavaScript reacts to user actions:

- Mouse clicks
- Keyboard input
- Page load events

👉 Enables responsive and interactive applications.

11. Real-Time Applications

JavaScript supports real-time features:

- Chat applications
- Live notifications
- Online gaming

👉 WebSockets and event handling enable real-time communication.

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12. Easy Integration with HTML and CSS

JavaScript works seamlessly with:

- HTML for structure
- CSS for styling

👉 Together, they form the core of modern web development.

13. Community Support

JavaScript has:

- Large developer community
- Extensive documentation
- Tutorials and forums

👉 Easy problem-solving and learning support.

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14. Mobile App Development

Using frameworks like:

- React Native
- Ionic

👉 JavaScript can be used to build cross-platform mobile apps.

15. Game Development

JavaScript supports:

- 2D and 3D games
- Canvas and WebGL

👉 Suitable for browser-based games.

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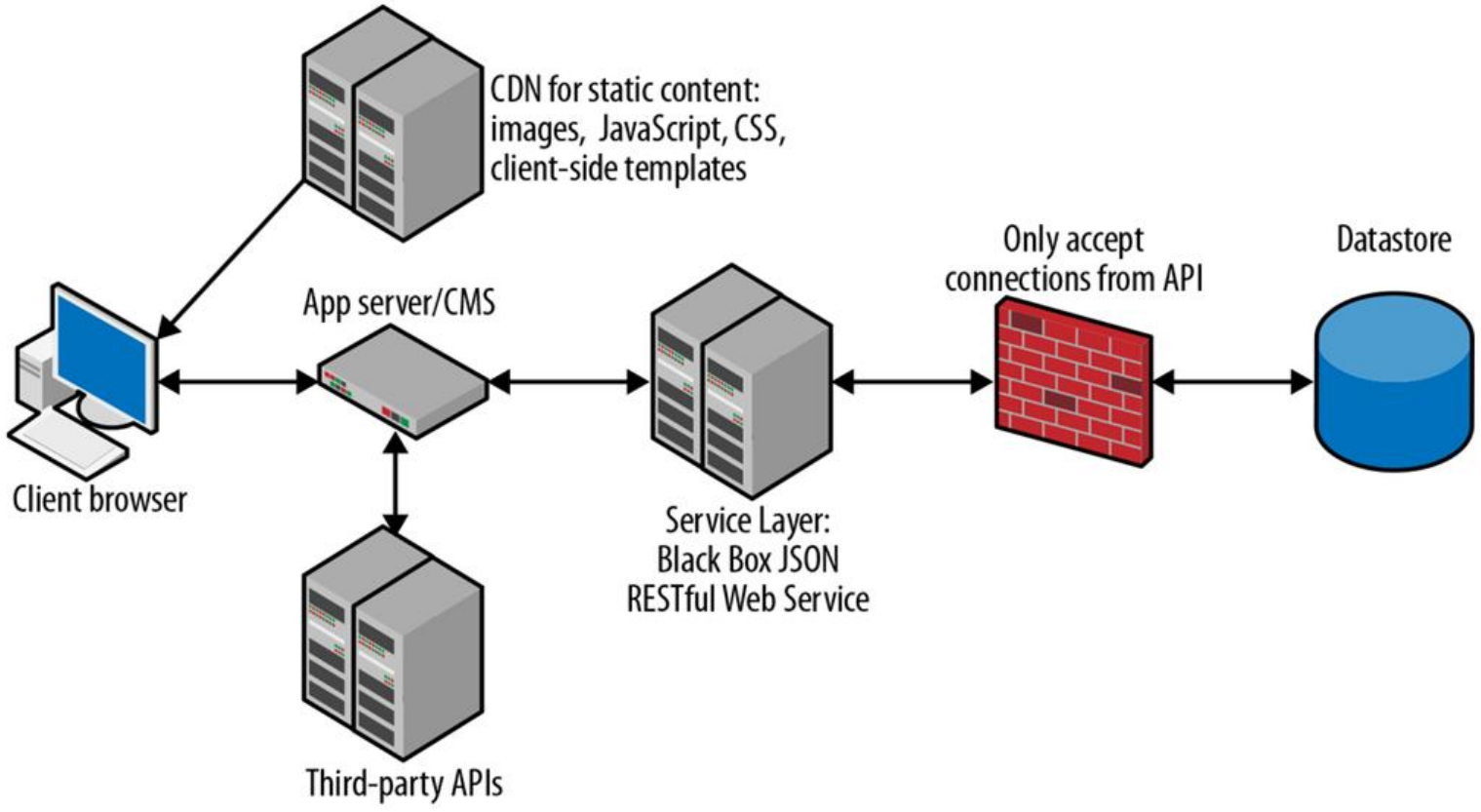
16. Continuous Evolution

JavaScript is continuously improving:

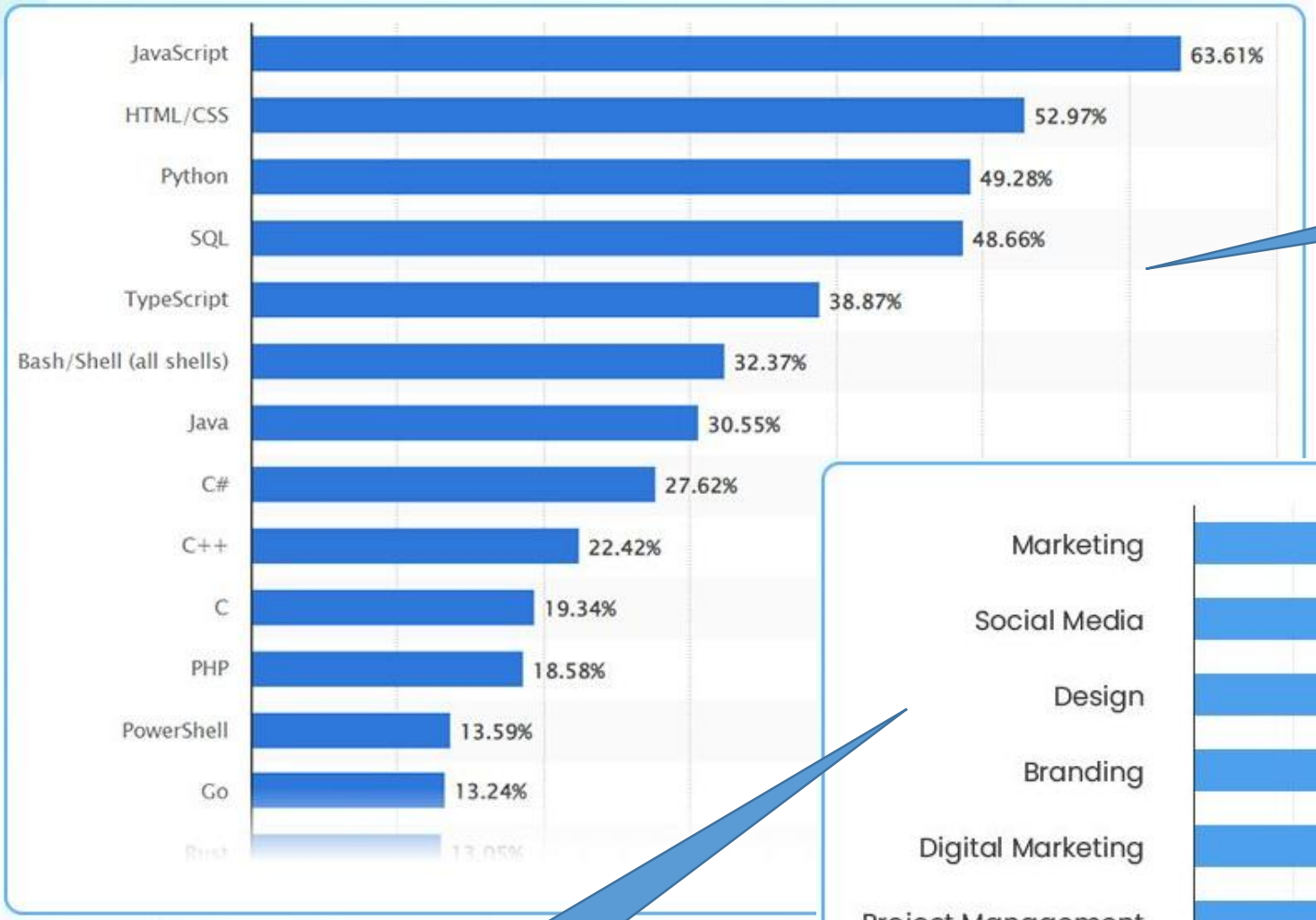
- New features added in ES6 and later versions
- Better performance and security

👉 Keeps the language modern and powerful.

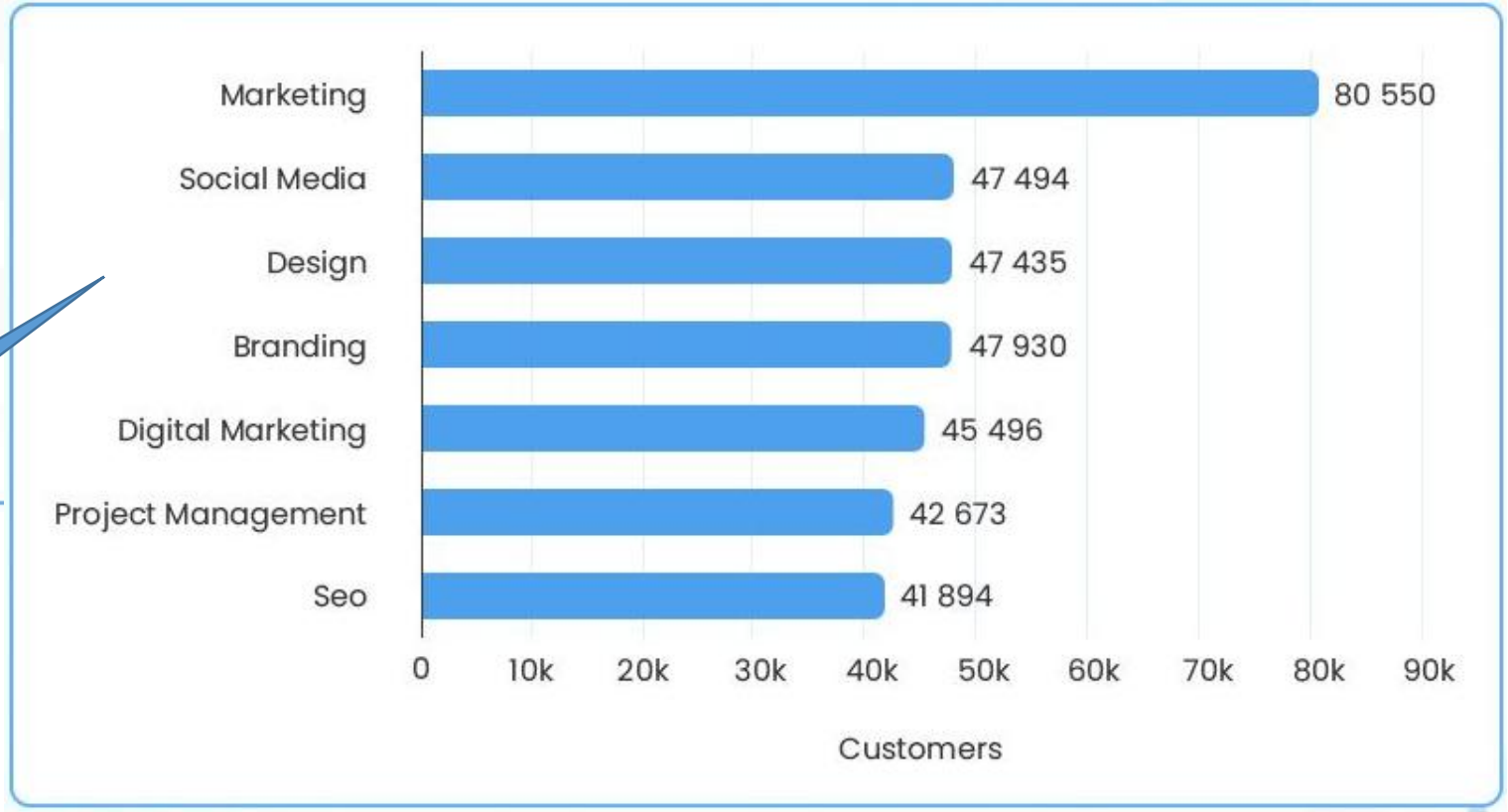
Anatomy of a Typical Modern JavaScript App



- Data store
- A virtual private network (VPN) or firewall (to protect the data store from unauthorized access)
- A black box JSON RESTful Web Service Layer
- Various third-party APIs
- An app server/content management system (CMS) to route requests and deliver pages to the client
- A static content deliver network (CDN) for cached files (like images, JavaScript, CSS, and client-side templates)
- The client (browser)



Most used programming language



Top industries using JavaScript